Air Pollution and Asthma Burden in the United States: A County-Level Analysis of Air Quality Index (AQI) and Emergency Department (ED) Visits

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Abstract

Background: Air pollution is a well-established trigger of asthma exacerbations; however, associations observed at the population level may be influenced by geographic confounding. This study evaluates if air pollution exposure is independently associated with asthma-related emergency department (ED) burden across U.S. counties.

Methods: This cross-sectional ecological study analyzed county-level data from 2023, comprising 528 observations representing 264 unique U.S. counties[1] across 13 states. The analytical dataset was constructed by merging county-level air quality data from the U.S. Environmental Protection Agency (EPA) with asthma emergency department (ED) visit data from the Centers for Disease Control and Prevention using common geographic identifiers (state, county, year). Each county contributed up to two observations corresponding to sex-specific asthma outcomes, resulting in a county-by-sex structure. Counties lacking complete data across sources were excluded via complete-case analysis.

Air pollution exposure was measured using an annual weighted Air Quality Index (AQI), calculated by combining AQI category midpoints with the frequency of days in each category, and further categorized into low, moderate, and high levels based on tertiles of the proportion of unhealthy AQI days. Asthma burden was defined as a binary indicator of elevated emergency department (ED) visit rates using sex-specific national thresholds. Bivariate associations between air pollution measures and asthma burden were assessed using Pearson chi-square, likelihood ratio, and Cochran–Armitage trend tests, with stratified analyses conducted using Cochran–Mantel–Haenszel methods. Multivariable logistic regression models were used to estimate adjusted associations, with state included as a key confounder to account for geographic variation, and model selection guided by likelihood ratio tests and Akaike Information Criterion (AIC). Model diagnostics included evaluation of residuals, goodness-of-fit testing (Pearson and Hosmer–Lemeshow), and assessment of influential observations using Cook's distance, with sensitivity analyses performed by excluding influential observations to assess the robustness of estimated associations.

Results: Unadjusted analyses demonstrated a statistically significant inverse association between AQI category and asthma burden ($\chi^2 = 22.61$, $p < 0.001$), with higher AQI categories showing lower proportions of elevated asthma burden. However, this counterintuitive pattern was explained by geographic confounding; after adjustment for state, the association was no longer significant ($p = 0.398$), and effect estimates shifted toward the null. In multivariable logistic regression models, state-level variation emerged as a dominant predictor of asthma burden. When AQI was analyzed as a continuous variable, it showed a positive association with asthma risk in adjusted analyses, although this effect was sensitive to model specification and became stronger and statistically significant after removal of influential observations. Predicted probabilities further demonstrated a consistent increase in asthma risk with increasing AQI across states, suggesting that higher air pollution increases asthma risk, even though there are differences between states.

Conclusion: The apparent inverse association between AQI and asthma burden in unadjusted county-level analyses was driven by geographic confounding. After appropriate adjustment and sensitivity analysis, higher air pollution was associated with increased asthma risk. However, substantial differences across states persisted, indicating that both environmental exposure and geographic factors contribute to asthma burden. These findings highlight the importance of careful model specification and accounting for geographic

[1] Unique counties represent distinct U.S. counties (n = 264); each county contributes two observations corresponding to sex-specific asthma outcomes.

variation in ecological analyses of environmental health data.

Keywords: Air pollution; AQI; asthma; logistic regression; ecological study; confounding

1. Introduction

Asthma is a chronic respiratory disease that represents a major public health burden in the United States. Current estimates indicate that approximately 23 million adults are affected, and the condition contributes to nearly 1.6 million emergency department (ED) visits annually (Centers for Disease Control and Prevention [CDC], 2023; Asthma and Allergy Foundation of America [AAFA], n.d.). These visits reflect acute exacerbations, which are often triggered by environmental exposures, particularly air pollution (Guamieri & Balmes, 2014). Understanding the relationship between environmental risk factors and asthma outcomes in therefore critical for both public health policy and clinical management.

Air Pollution is commonly measured using the Air Quality Index (AQI), a standardized metric developed by the Environmental Protection Agency (EPA). The AQI integrates multiple pollutants, including particulate matter (PM2.5 and PM10), ozone, nitrogen dioxide (NO_2), and carbon monoxide (CO), into a single scale that reflects overall air quality and associated health risks (EPA, n.d.). Higher AQI values correspond to poorer air quality and greater potential for adverse health outcomes. Importantly, AQI captures both the frequency and severity of pollution exposure over time, thereby making it a useful measure for population-level analyses.

Previous studies have demonstrated positive associations between air pollution and asthma-related outcomes, with exposure to pollutants such as PM2.5 and ozone linked to increased asthma exacerbations, hospitalizations, and ED visits (Guarnieri & Balmes, 2014; Orellano et al., 2017). However, these relationships are complex and often confounded by geographic, socioeconomic, and healthcare-related factors. This complexity is reflected in the present analysis, where unadjusted results showed a counterintuitive inverse association between AQI and asthma burden. Counties with higher AQI appeared to have lower rates of elevated asthma ED visits, a pattern that is inconsistent with established biological evidence. After accounting for state-level variation, this association disappeared, indicating that the initial finding was driven by geographic confounding. This supports the idea that differences in healthcare access, urbanization, and regional characteristics, such as those described in prior studies, can distort population-level associations. For example, counties with higher pollution levels may also have better healthcare infrastructure, leading to lower observed emergency department burden despite higher exposure, whereas counties with lower pollution may experience higher ED utilization due to limited access to care. These findings are consistent with the ecological framework described by Morgenstern (1998) and highlight how underlying geographic differences can mask the true relationship between air pollution and asthma outcomes.

From a statistical perspective, studies that use group-level data (like counties) are more likely to be affected by confounding and bias. Common methods such as contingency tables and logistic regression assume that observations are independent, but this assumption may not hold when data are grouped by location. Counties within the same state may be more similar to each other than to counties in other states. If this is not accounted for, the results can be misleading or even show the wrong direction of association (Agresti, 2013).

This study aims to examine whether air pollution exposure is associated with higher asthma-related emergency department (ED) visit rates at the county level across the United States. Exposure is measured using both continuous and categorical AQI to capture different aspects of air pollution. The analysis accounts for state-level differences in healthcare access, policy, and environmental conditions. By integrating multiple measures of exposure and applying standard categorical data analysis methods, this study seeks to provide a more accurate and nuanced understanding of the relationship between air pollution and asthma burden.

1.1 Research Question

Is air pollution exposure, measured using both continuous (annual weighted AQI) and categorical indicators, associated with elevated asthma-related emergency department (ED) visit rates across U.S. counties after accounting for state-level variation?

1.2 Hypotheses

The null hypothesis (H_0) is that air pollution exposure is not associated with elevated asthma burden after adjusting for state-level effects ($\beta_{\text{AQI}} = 0$).
The alternative hypothesis (H_1) is that air pollution exposure is associated with elevated asthma burden after adjustment ($\beta_{\text{AQI}} \neq 0$).

2. Methods

2.1 Study Design and Data Sources

This study employs a cross-sectional ecological design, with the county as the primary unit of analysis. The analytic dataset includes 528 observations representing 264 unique U.S. counties across 13 states for the year 2023. Each county contributed up to two observations corresponding to sex-specific asthma outcomes, resulting in a county-by-sex data structure. This design allows for the examination of associations between environmental exposure and health outcomes at the population level but does not support causal inference at the individual level.

Two primary datasets were used. The first, obtained from the U.S. Environmental Protection Agency (EPA), provides county-level air quality data, including the number of days in each AQI category and pollutant-specific contributions. The second, obtained from the Centers for Disease Control and Prevention (CDC), contains age-adjusted asthma-related emergency department (ED) visit rates per 10,000 population, stratified by sex. These rates serve as a standardized measure of asthma burden, enabling comparisons across geographic regions.

The datasets were merged using common identifiers, including state, county, and year. Counties lacking complete or matched data across sources were excluded via complete-case analysis. The final analytic dataset consisted of 528 observations representing 264 unique counties. Each county contributed two observations corresponding to sex-specific asthma outcomes, resulting in a county-by-sex data structure. While this approach ensures consistency across variables, it may introduce selection bias if excluded counties differ systematically from those included. A summary of the analytic dataset is presented in Table 1.

Table 1. Summary of the Merged County-Level Dataset (United States, 2023)

Number of Observations (County-Level)	Number of States	Number of Counties	Sex Categories	Year
528	13	264	2	2023

2.2 Variable Construction

Air pollution exposure was measured using multiple indicators to capture different aspects of environmental risk. The primary continuous measure, annual weighted AQI, reflects both the frequency and severity of air pollution over the course of a year. This variable was calculated by assigning midpoint values to each AQI category and computing a weighted average based on the number of days in each category:

$$\text{AQI}_{\text{weighted}} = \frac{\sum(\text{Midpoint}_\text{i} \ \times \ \text{Days}_\text{i})}{\text{Total Days}}$$

where midpoint values correspond to standard AQI categories, based on the U.S. Environmental Protection Agency (EPA) Air Quality Index classification system (e.g., 25 for “Good,” 75 for “Moderate,” 125 for “Unhealthy for Sensitive Groups”).

A second exposure measure, proportion of unhealthy days (prop_unhealthy), was calculated as:

$$\text{prop}_{\text{unhealthy}} = \frac{\text{USG} + \text{Unhealthy} + \text{Very Unhealthy} + \text{Hazardous Days}}{\text{Total AQI Days}}$$

This variable captures the frequency of days with potentially harmful air quality.

To facilitate categorical analysis, counties were further classified into low, moderate, and high AQI categories based on tertiles of the prop_unhealthy distribution. These categories are ordinal and approximately balanced in size. In addition, a pollutant category (nominal) variable was created to identify the dominant pollutant in each county, defined as the pollutant contributing the greatest number of AQI days (PM2.5, ozone, PM10, CO, NO_2, or mixed).

The outcome variable, asthma ED status, is a binary indicator of whether a county’s asthma-related emergency department visit rate exceeds sex-specific national thresholds. Counties were classified as “*elevated*” if rates exceeded 27.1 per 10,000 for males or 32.3 per 10,000 for females. Formally, let Y_i denote the asthma ED status for county i, where $Y_i = 1$ if asthma burden is elevated and $Y_i = 0$ otherwise. We assume $Y_i \sim \text{Bernoulli}(p_i)$, where $p_i = P(Y_i = 1)$ represents the probability that county i has elevated asthma burden. The probability mass function of the Bernoulli distribution is:

$$P(Y_i = y_i) = p_i^{y_i}(1 - p_i)^{1-y_i}, y_i \in \{0,1\}$$

2.3 Statistical Analysis

The analytical approach is grounded in the framework of categorical data analysis, as described in Agresti's An Introduction to Categorical Data Analysis. Because the outcome variable, elevated asthma emergency department (ED) status is binary, logistic regression was used as the primary modeling approach. Logistic regression is a generalized linear model (GLM) for Bernoulli-distributed data that relates a binary outcome to explanatory variables through the logit link function:

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_P X_p$$

where $P(Y=1)$represents the probability that a county has elevated asthma burden. Model coefficients are interpreted on the log-odds scale, and exponentiating coefficients yields odds ratios (ORs), which quantify the multiplicative change in odds associated with each predictor.

Bivariate Analysis. Unadjusted associations between air pollution exposure and asthma burden were evaluated using contingency table methods. Specifically, Pearson chi-square tests and likelihood ratio (G^2) tests were used to assess independence between categorical variables, while the Cochran–Armitage trend test was applied to evaluate monotonic trends for ordinal exposure variables such as AQI category. The chi-square test compares observed and expected counts under the assumption of independence and is given by:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

where O represents observed counts and E represents expected counts.

The likelihood ratio test provides an alternative measure based on likelihood theory and is defined as:

$$G^2 = 2 \sum O \log\left(\frac{O}{E}\right)$$

Together, these tests evaluate whether the distribution of asthma burden differs across levels of air pollution exposure.

Multivariable Modeling. Multivariable analysis was conducted using logistic regression models with air pollution exposure as the primary predictor. State was included as a key covariate to account for geographic variation and potential confounding, consistent with the study's emphasis on state-level effects. The primary model can be written as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 AQI_i + \sum_{k=1}^{K} \gamma_k State_{ik}$$

where p_i is the probability of elevated asthma burden for county i, β_0 is the intercept, β_1 represents the effect of AQI, γ_k are coefficients for state-level indicators, and $State_{ik}$ are dummy variables for each state.

Interpretation of Parameters. The coefficient for AQI (β_1) represents the change in log-odds of elevated asthma burden associated with a one-unit increase in AQI. Exponentiating this coefficient yields the odds ratio:

$$OR = e^{\beta_1}$$

- If OR > 1: AQI increases asthma risk
- If OR < 1: AQI decreases asthma risk
- If OR = 1: No Association

Estimation. Model parameters were estimated using maximum likelihood estimation (MLE) under the assumption of independent observations. Given the Bernoulli formulation of the outcome, the likelihood for all observations is:

$$L(\boldsymbol{\beta}) = \prod_{i=1}^{n} p_i^{y_i} (1-p_i)^{1-y_i}$$

where $p_i = P(Y_i = 1)$ represents the probability that county i has elevated asthma burden. The corresponding log-likelihood is:

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^{n} [y_i \log(p_i) + (1-y_i)\log(1-p_i)]$$

Parameter estimates were obtained by maximizing the log-likelihood.

2.4 Model Assumptions and Diagnostics

The validity of the logistic regression model relies on several key assumptions. In this analysis, each assumption was evaluated using both theoretical considerations and empirical diagnostics presented in the results.

Independence of Observations. The logistic regression model assumes that observations are independent. In this study, each observation corresponds to a county; however, this assumption may be violated due to geographic clustering, as counties within the same state may share similar environmental, healthcare, and socioeconomic characteristics.

This concern is supported by the results, where adjustment for state-level effects substantially altered the estimated association between AQI and asthma burden (Table 12), indicating the presence of within-state dependence. To partially address this, state indicators were included as fixed effects in the model. However, this approach does not fully capture correlation between neighboring counties, and therefore the independence assumption may still be violated. This limitation is further discussed in the context of hierarchical and spatial modeling approaches.

Appropriateness of the Logit Link Function. Logistic regression assumes that the logit link function appropriately relates the predictors to the probability of the outcome. That is,

$$\text{logit}\,(p_i) = \log\left(\frac{p_i}{1 - p_i}\right)$$

is assumed to be a linear function of the predictors.

Model fit was evaluated using goodness-of-fit tests, including the Hosmer–Lemeshow test (Table 22), which was not statistically significant (p = 0.859), indicating no evidence of lack of fit. Additionally, deviance and Pearson residuals (Figures 5–6) did not show systematic departures from model assumptions. Together, these results support the appropriateness of the logit link for this analysis.

Linearity in the Logit for Continuous Predictors. For continuous predictors such as annual weighted AQI, logistic regression assumes a linear relationship between the predictor and the log-odds of the outcome.

This assumption was assessed by examining residual plots and model specification. The residuals versus fitted values plot (Figure 7) did not indicate strong systematic curvature, suggesting that the linearity assumption is reasonable. However, given the relatively narrow range of AQI values and the lack of statistical significance in the continuous AQI model (Table 14), it is possible that nonlinearity or threshold effects are present but not detected in this specification. To address this, AQI was also modeled as an ordinal categorical variable, allowing for a more flexible representation of exposure without imposing a strict linearity assumption.

Absence of Multicollinearity. Logistic regression assumes that predictors are not highly correlated, as multicollinearity can inflate standard errors and destabilize coefficient estimates.

This assumption was evaluated using variance inflation factors (VIF), as shown in Table 20. All VIF values were below 2, indicating no evidence of problematic multicollinearity. This suggests that the estimated regression coefficients are stable and interpretable.

Influence and Outliers. Although not always listed as a formal assumption, logistic regression is sensitive to influential observations. This was assessed using Cook's distance (Tables 21 and 24), which identified several observations with high influence on model estimates.

Sensitivity analyses demonstrated that removal of these observations improved model fit and predictive performance (as reflected in changes in AIC and ROC curves) but did not materially alter the primary conclusion that AQI is not independently associated with asthma burden after adjustment for state.

2.5 Incorporation of Geographic Effects

As discussed in the model assumptions, the independence of observations may be violated due to geographic clustering of counties within states. To address this, state-level indicators were included as fixed effects in the logistic regression model, allowing each state to have its own intercept and controlling for between-state confounding. As demonstrated in the adjusted analyses (Table 12) and multivariable model (Table 17), inclusion of state effects substantially alters the estimated association between AQI and asthma burden, indicating the importance of geographic variation.

While including state fixed effects helps control for confounding, it adds many parameters to the model, which can make it harder to interpret and less efficient. It also accounts for differences between states without directly modeling how observations within the same state are related.

A natural extension is a multilevel (hierarchical) modeling framework, in which counties are nested within states and state is treated as a random effect:

$$\log\left(\frac{P(Y_{ij}=1)}{1-P(Y_{ij}=1)}\right) = \beta_0 + \beta_1 X_{ij} + u_{state,j}, u_{state,j} \sim N(0,\sigma^2)$$

where j indexes states. This formulation allows for partial pooling across states and provides a more efficient representation of geographic variability.

In addition, spatial modeling approaches may be useful, since air pollution and healthcare access tend to follow geographic patterns. These models account for similarities between neighboring counties, rather than assuming all observations are independent, as in standard logistic regression.

Overall, while using state fixed effects helps control for confounding, the results, especially the weakening of the AQI effect after adjustment, show that geographic structure plays an important role. More advanced approaches, such as hierarchical and spatial models, could better capture these patterns, but they were beyond the scope of this course.

3. Results

3.1 Descriptive Statistics

Descriptive analyses (Tables 2 and 3) showed that AQI categories were relatively balanced (36% low, 30% moderate, 34% high), and 58% of counties had elevated asthma burden. Continuous variables exhibited substantial variability, with annual weighted AQI ranging from 25 to 80 (mean 41.6, SD 9.2) and asthma ED visit rates ranging from 5 to 82 per 10,000. The proportion of unhealthy AQI days was low overall (mean = 2%). Figure 2 shows the distribution of continuous variables. Asthma ED visit rates exhibit substantial variability with several high-value outliers, while annual weighted AQI is more centrally distributed. The proportion of unhealthy days is right-skewed, with most counties experiencing low but non-zero exposure.

Table 2. Descriptive Summary of Categorical Variables (N = 528)

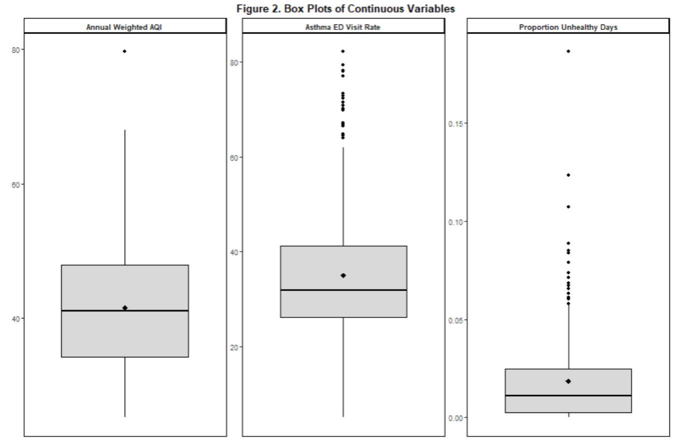
Characteristic	N = 528[1]
AQI Category	
Low	192 (36%)
Moderate	158 (30%)
High	178 (34%)
Pollutant Category	
Ozone	208 (39%)
PM2.5	304 (58%)
PM10	16 (3.0%)
Sex	
Female	264 (50%)
Male	264 (50%)
Asthma ED Status	
Not Elevated	224 (42%)
Elevated	304 (58%)

[1] n (%)

Table 3. Summary of Continuous Variables (N = 528)

Characteristic	N = 528
Proportion Unhealthy Days	
Mean (SD)	0.02 (0.02)
Median	0.01
Min – Max	0.00 – 0.19
Asthma ED Visit Rate	
Mean (SD)	35.01 (12.89)
Median	31.85
Min – Max	5.10 – 82.10
Annual Weighted AQI	
Mean (SD)	41.56 (9.15)
Median	41.06
Min – Max	25.14 – 79.66

Figure 2. Box Plots of Continuous Variables



3.2 Bivariate Analysis

Contingency table analyses revealed a strong inverse association between AQI category and asthma burden, with higher AQI associated with a lower prevalence of elevated asthma burden (Table 4).

Table 4. Association Between AQI Category and Asthma ED Status

	Asthma_ED_Status		
AQI Category	Not Elevated	Elevated	**Total**
AQI_Category			
Low	60 (31%)	132 (69%)	192 (100%)
Moderate	65 (41%)	93 (59%)	158 (100%)
High	99 (56%)	79 (44%)	178 (100%)
Total	224 (42%)	304 (58%)	528 (100%)

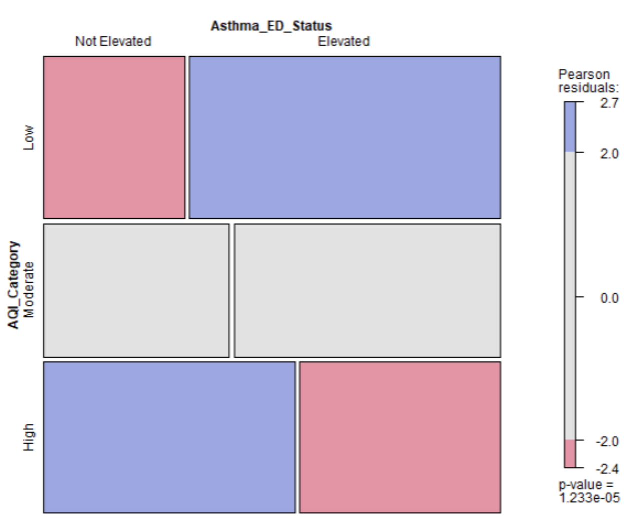
Statistical tests confirmed this association, including the Pearson chi-square test ($\chi^2 = 22.61, p < 0.001$), likelihood ratio test ($G^2 = 22.74, p < 0.001$), and Cochran–Armitage trend test ($Z = 4.73, p < 0.001$) (Combined Table: Tests of Association between AQI Category and Asthma ED Status: Derived from Tables 5–7).

Tests of Association Between AQI Category and Asthma ED Status

Test	Statistic	Degrees of Freedom	p-value
Chi-square	22.61	2	<0.001
Likelihood Ratio (G^2)	22.74	2	<0.001
Cochran–Armitage Trend	4.73	NA	<0.001

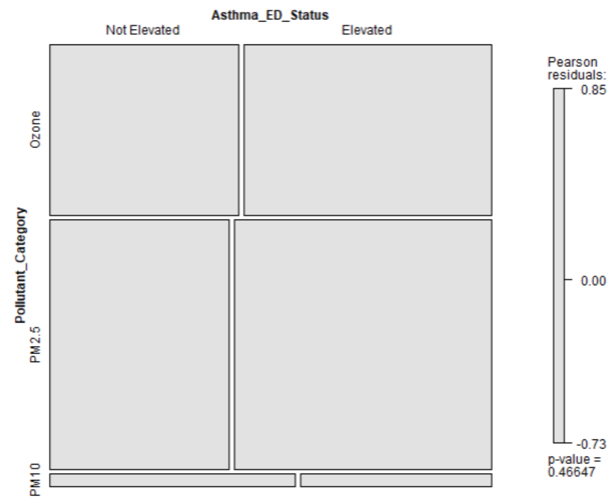
Mosaic plot further illustrates deviations from independence between AQI category and asthma burden (Figure 3). In particular, low AQI counties showed a higher number of elevated asthma cases than expected under independence, while high AQI counties showed fewer elevated cases than expected. These deviations indicate that the observed distribution differs from what would be expected if AQI and asthma burden were unrelated, reinforcing the inverse pattern seen in the contingency table.

Figure 3. Mosaic Plot of AQI Category and Asthma ED Status



Pollutant Category Analysis. Analyses of dominant pollutant category showed no clear or consistent association with asthma burden. Counties dominated by PM2.5 and ozone exhibited similar proportions of elevated asthma cases, while PM10 counties were few and showed no discernible pattern. Pearson chi-square tests indicated no statistically significant association between pollutant category and asthma ED status (p > 0.05), and standardized residuals showed minimal deviation from expected counts across categories. Mosaic plot (Figure 4) further supported these findings, showing no substantial deviations from independence across pollutant categories. Overall, these results suggest that asthma burden does not differ meaningfully by dominant pollutant type, indicating that overall pollution exposure may be more relevant than the specific pollutant driving AQI levels.

Figure 4. Mosaic Plot of Pollutant Category and Asthma ED Status



3.3 Stratified and Adjusted Analysis

Stratified analyses by sex (Tables 10–11) showed that the inverse association between AQI category and asthma burden persisted in both males and females, with higher AQI categories consistently associated with a lower proportion of elevated asthma cases. The Cochran–Mantel–Haenszel (CMH) test confirmed that this association remained statistically significant after adjusting for sex ($\chi^2 = 22.59$, $p < 0.001$), indicating that sex did not explain the observed relationship.

As shown in Table 13, the crude odds ratios, representing the unadjusted relationship between AQI and asthma burden without accounting for other factors, indicate a strong inverse association, with counties in the high AQI category having substantially lower odds of elevated asthma burden compared to low AQI counties (OR = 0.36, 95% CI: 0.24–0.55, $p < 0.001$). In contrast, the adjusted odds ratios, estimated after accounting for differences across states, show that this association reverses direction and is no longer statistically significant (OR = 1.41, 95% CI: 0.72–2.78, $p = 0.311$). A similar weakening is observed for moderate AQI levels. This change from the crude to the adjusted results suggests that the initial inverse relationship was influenced by differences between states rather than the true effect of air pollution.

In contrast, adjustment for state (Table 12) substantially altered the results. The CMH test was no longer statistically significant ($\chi^2 = 1.84$, $p = 0.398$), and the previously observed inverse association disappeared. This finding indicates that the initial AQI–asthma relationship was largely driven by geographic differences across states, suggesting strong confounding by state-level factors such as healthcare access, environmental conditions, and population characteristics.

Table 12. Cochran–Mantel–Haenszel Test Adjusted for State

CMH Chi-Square	Degrees of Freedom	p-value
1.84	2	0.398

3.4 Logistic Regression Models

Unadjusted logistic regression models produced mixed results depending on how air pollution exposure was specified (Table 15). When modeled continuously, annual weighted AQI was not significantly associated with elevated asthma burden (OR ≈ 0.99, $p = 0.477$). However, when modeled categorically, higher AQI categories were associated with lower odds of elevated asthma burden, with counties in the high AQI group showing substantially reduced odds compared to low AQI counties (OR ≈ 0.36, $p < 0.001$). This apparent inverse association is consistent with the bivariate findings and likely reflects underlying geographic differences rather than a true protective effect of air pollution.

Table 15. Logistic Regression Results: AQI Category (Unadjusted Model)

Term	Coefficient	Std. Error	z value	p-value	2.5% (log-odds)	97.5% (log-odds)	OR	2.5% (OR)	97.5% (OR)
(Intercept)	0.7885	0.1557	5.0640	<0.001	0.4885	1.1001	2.2000	1.6299	3.0044
AQI_CategoryModerate	-0.4302	0.2245	-1.9169	0.055	-0.8720	0.0090	0.6503	0.4181	1.0090
AQI_CategoryHigh	-1.0141	0.2168	-4.6777	<0.001	-1.4433	-0.5926	0.3627	0.2362	0.5529

After adjusting for state (Table 17), the interpretation changed substantially. In models including annual weighted AQI and state, AQI showed a small positive association with asthma burden (OR ≈ 1.03, $p = 0.026$), indicating that higher pollution levels were associated with increased asthma risk when comparisons were made within states. However, this effect became non-significant after further adjustment for pollutant category, while state remained a strong and consistent predictor across all models.

Overall, these results indicate that the initial inverse association observed in unadjusted analyses was driven by confounding due to state-level differences. Once geographic variation was properly accounted for, the relationship between AQI and asthma burden shifted direction and weakened, highlighting the importance of model specification and adjustment for confounding factors.

3.5 Model Evaluation and Selection

Model evaluation began by assessing the contribution of AQI to model fit in unadjusted analyses. As shown in Table 16, the likelihood ratio test comparing the null model (no predictors) to a model including AQI category was statistically significant ($\chi^2 = 22.74$, df = 2, $p < 0.001$), indicating that AQI category is associated with asthma ED status and improves model fit in the unadjusted setting. However, this improvement reflects the same inverse association observed in earlier analyses and may be influenced by underlying geographic differences.

Table 16. Likelihood Ratio Test: Contribution of AQI Category

Comparison	Chi-Square	df	p-value
AQI Category vs Null Model	22.74	2	<0.001

Further comparison of multivariable logistic regression models using Akaike Information Criterion (AIC) and likelihood ratio tests (Table 18) showed that differences in model fit were small across specifications. The base model including annual weighted AQI and state had the lowest AIC (≈ 541.5), indicating the best overall fit. Adding pollutant category resulted in only a negligible increase in AIC and did not significantly improve model fit (Δ Deviance = 1.50, $p = 0.472$). Similarly, replacing continuous AQI with categorical AQI resulted in a worse-fitting model (higher AIC ≈ 546.6), suggesting that categorizing AQI reduces model performance. Overall, these results indicate that the base model provides the best balance of fit and parsimony.

Model	AIC	Comparison	ChiSq	df	p
Base (Annual Weighted AQI + State)	541.5066	—	—	—	—
Expanded (+ Pollutant)	544.0061	Base vs Expanded	1.5	2	0.472
Alternative (AQI Category + State)	546.6419	Base vs Alternative	-3.14	1	NA

Table 18. Model Comparison (AIC and Likelihood Ratio Tests)

Interaction Effects. Potential interaction effects were evaluated using likelihood ratio tests (Table 19) to determine whether the association between AQI and asthma burden varied across states or pollutant categories. There was no statistically significant evidence of interaction between AQI and state (p = 0.599 for AQI category; p = 0.078 for continuous AQI), nor between AQI and pollutant category (p = 0.521). These findings indicate that the effect of AQI on asthma burden does not vary meaningfully across states or pollutant types. Instead, state appears to influence the overall level of asthma burden (confounding), rather than modifying the relationship between AQI and asthma.

Table 19. Likelihood Ratio Tests for Interaction Effects

Interaction	Chi-Square	df	p-value
AQI Category × State	16.87	19	0.599
Annual Weighted AQI × State	19.45	12	0.078
Annual Weighted AQI × Pollutant Category	1.30	2	0.521

Variable Selection. Variable selection procedures were carried out using forward, backward, and stepwise methods (Table 20) to identify the most appropriate set of predictors for the model. Across all three approaches, the same final model was selected, which included annual weighted AQI and state. This means that no matter how the model was built, whether starting with no variables, all variables, or adding and removing step by step, the result was consistent.

The AIC values were identical across all methods (≈ 541.51), indicating that this model provides the best balance between goodness of fit and simplicity. Importantly, pollutant category was not selected in any of the procedures, suggesting that it does not improve the model's ability to explain asthma burden beyond what is already captured by AQI and state. Overall, the agreement across all selection methods shows that the final model is stable and reliable. It also reinforces that state and overall air pollution (measured by AQI) are the key factors associated with asthma burden in this analysis, while pollutant type does not add meaningful additional information.

Table 20. Variable Selection Results Using Stepwise Procedures

Selection Method	Final Model	AIC
Forward Selection	Asthma_ED_Status ~ State + Annual_Weighted_AQI	541.51
Backward Selection	Asthma_ED_Status ~ Annual_Weighted_AQI + State	541.51
Stepwise Selection	Asthma_ED_Status ~ Annual_Weighted_AQI + State	541.51

3.6 Final Model Specification

Based on model comparison and variable selection results, the final model includes annual weighted AQI and state as predictors of asthma ED status. The final logistic regression model is given by:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 AQI_i + \sum_{k=1}^{K} \gamma_k State_{ik}$$

where $p_i = P(Y_i = 1)$ is the probability of elevated asthma burden for county i, β_0 is the intercept, β_1 represents the effect of AQI, γ_k are coefficients for state-level indicators, and $State_{ik}$ are dummy variables for each state.

In this model, AQI is included as a continuous measure to capture changes in pollution exposure, while state is included as a set of indicator variables to account for geographic differences across regions. This specification allows the model to estimate the association between AQI and asthma burden while controlling state-level factors.

3.7 Model Diagnostics

Model diagnostics (Tables 21–26) indicated that the fitted logistic regression models were appropriate and provided a good representation of the data. There was no evidence of multicollinearity, with all variance inflation factor (VIF) values close to 1 and well below commonly used thresholds (Table 21), indicating that predictors contribute independent information and that model estimates are stable.

Table 21. Variance Inflation Factors (VIF) for Final and Expanded Models

Variable	VIF (Final Model)	Df.x	GVIF^(1/(2*Df)).x	VIF (Expanded Model)	Df.y	GVIF^(1/(2*Df)).y
Annual_Weighted_AQI	1.21	1	1.10	1.83	1	1.35
State	1.21	12	1.01	1.90	12	1.03
Pollutant_Category	NA	NA	NA	2.10	2	1.20

Residual diagnostics supported adequate model fit. As shown in Figures 5 and 6, both Pearson and deviance residuals were centered around zero, with most values falling within the expected range (±2). The plots display the characteristic two-band structure expected for a binary outcome, where one band corresponds to counties with elevated asthma burden and the other to counties without elevated burden.

Figure 5: Pearson Residuals vs Observation Index

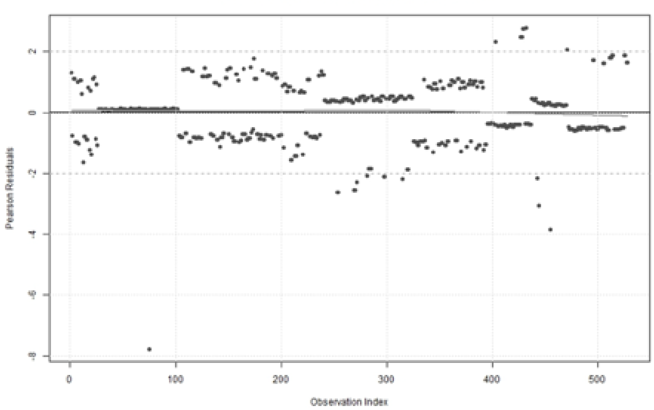
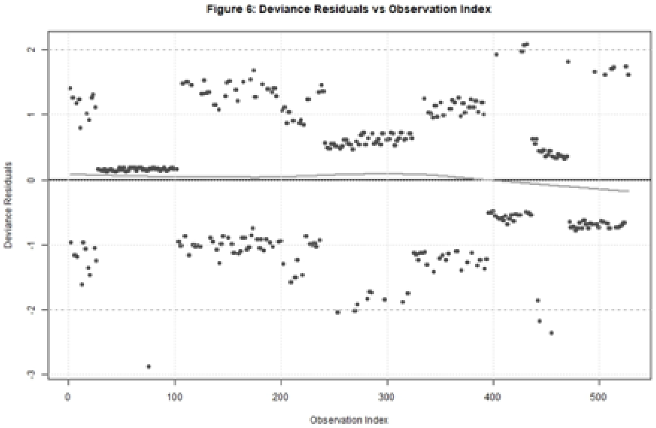
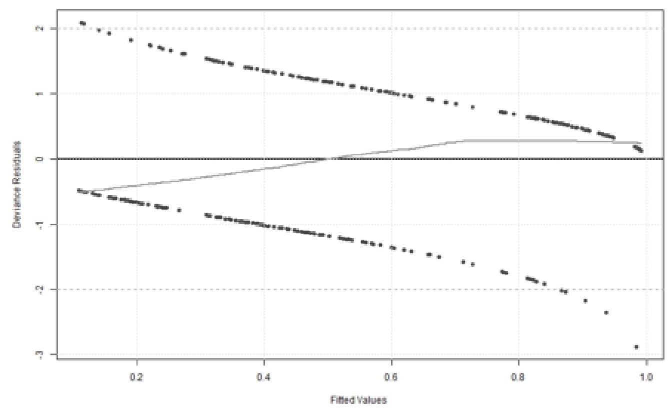


Figure 6: Deviance Residuals vs Observation Index



In the residuals versus fitted values plots (Figure 7), the points are spread across the range of predicted probabilities without any clear pattern or trend. The residuals remain centered around zero throughout, and the smoothing line is relatively flat with only slight curvature. This indicates that the model does not systematically overestimate or underestimate the outcome. While a few larger residuals are observed at higher fitted values, these are limited in number and do not appear to drive the overall results. Overall, these patterns suggest that the model provides a good fit to the data and that there is no strong evidence of model misspecification.

Figure 7: Residuals vs Fitted Values



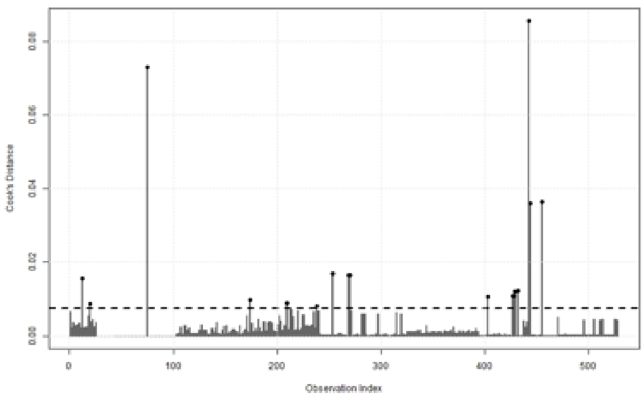
Influence diagnostics based on Cook's distance identified a small number of potentially influential observations (Table 22). However, these values were generally low, with only a few exceeding the conventional threshold (4/n) (Figure 8), and were dispersed across multiple states and outcome groups. This indicates that no single observation disproportionately influenced the model, and that the results are not driven by a specific subset of counties.

Table 22. Influential Observations Identified by Cook's Distance

Observations with Cook's Distance > 4/n

Observation	Cooks_Distance	Asthma_ED_Status	Annual_Weighted_AQI	State
442	0.0856	Not Elevated	35.98592	Rhode Island
75	0.0730	Not Elevated	29.77178	Florida
455	0.0365	Not Elevated	45.27397	South Carolina
444	0.0360	Not Elevated	29.18182	South Carolina
253	0.0169	Not Elevated	45.12363	New Jersey
254	0.0169	Not Elevated	45.12363	New Jersey
269	0.0166	Not Elevated	43.14404	New Jersey
270	0.0166	Not Elevated	43.14404	New Jersey
12	0.0157	Not Elevated	79.65753	Arizona
432	0.0124	Elevated	31.84932	Oregon
429	0.0121	Elevated	32.75623	Oregon
430	0.0121	Elevated	32.75623	Oregon
427	0.0108	Elevated	40.75342	Oregon
428	0.0108	Elevated	40.75342	Oregon
403	0.0106	Elevated	45.20548	Oregon
174	0.0099	Elevated	26.66667	Iowa
209	0.0090	Not Elevated	55.54795	Kansas
210	0.0090	Not Elevated	55.54795	Kansas
20	0.0087	Not Elevated	68.01370	Arizona
238	0.0082	Elevated	32.06997	Nebraska

Figure 8: Cook's Distance



Goodness-of-fit tests (Table 23) further supported model adequacy. The Pearson chi-square goodness-of-fit test was non-significant, and the Hosmer–Lemeshow test showed no evidence of lack of fit (p = 0.859). Additionally, there was a substantial reduction in deviance compared to the null model, indicating improved fit with the inclusion of predictors.

Table 23. Model Fit and Goodness-of-Fit Statistics

Measure	Statistic	df	p-value
Model Deviance	513.51	NA	NA
Null Deviance	719.80	NA	NA
Deviance Difference (LRT)	206.29	NA	NA
Pearson Chi-Square	512.35	514	0.512
Hosmer–Lemeshow Test	3.98	8	0.859

Contribution of AQI to Model Fit. The contribution of annual weighted AQI to the final model was formally evaluated using a likelihood ratio test comparing a state-only model to a model including both state and AQI (Table 24). The test was statistically significant (χ^2 = 5.04, df = 1, p = 0.025), indicating that adding AQI significantly improves model fit. This improvement is reflected in the reduction in deviance, suggesting that AQI explains additional variation in asthma ED status beyond what is captured by state alone. These findings demonstrate that, even after accounting for geographic differences, air pollution remains an important predictor of asthma burden. In other words, AQI is not simply acting as a proxy for state-level variation but provides independent explanatory value in the model.

Table 24. Contribution of Annual Weighted AQI to Model Fit (Likelihood Ratio Test)

Comparison	Test Statistic (χ^2)	df	p-value
State-only model vs State + Annual Weighted AQI model	5.04	1	0.025

Sensitivity Analysis. Sensitivity analyses were conducted by removing influential observations to assess the robustness of the results (Table 25). After their removal, model fit improved substantially, as indicated by lower AIC and deviance values. The estimated effect of AQI also increased (from 0.03 to 0.05) and became more statistically significant (p = 0.027 to p < 0.001), indicating a stronger association with asthma burden.

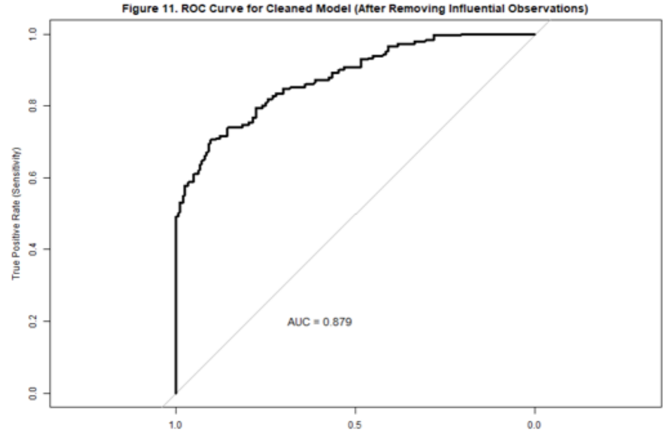
Table 25. Comparison of Final Model and Sensitivity Analysis Model (After Removing Influential Observations)

Model	Estimate	Std. Error	p_value	AIC	Deviance
Final Model	0.03	0.01	0.027	541.51	513.51
Cleaned Model	0.05	0.01	<0.001	434.49	406.49

These results suggest that a small number of influential counties were weakening the observed relationship in the original model. Importantly, the direction of the association remained the same, indicating that the findings are robust and not driven by a few extreme observations. Instead, removing these observations reveals a clearer and stronger underlying relationship between air pollution and asthma burden.

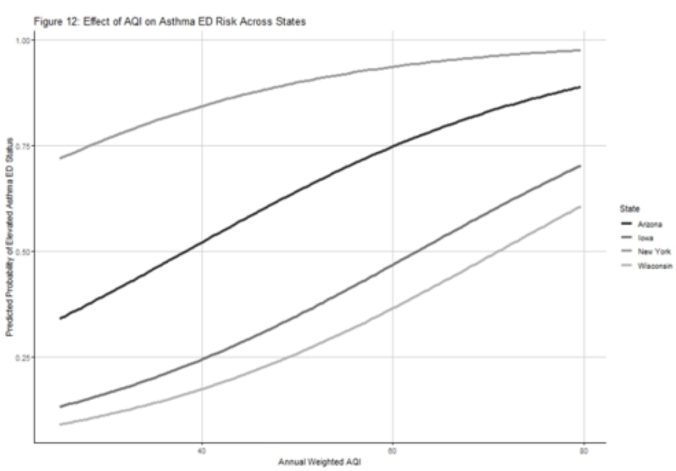
ROC Curve. Model discrimination was evaluated using the receiver operating characteristic (ROC) curve (Figure 8), which plots the trade-off between sensitivity and specificity across different classification thresholds. The model demonstrated good discriminatory ability (AUC ≈ 0.835), which improved further after removal of influential observations (AUC ≈ 0.879), indicating strong performance in distinguishing between elevated and non-elevated asthma burden.

Figure 11. ROC Curve for Cleaned Model (After Removing Influential Observations)



State Effects (Adjusted for AQI). Results from the adjusted model (Table 26) show that meaningful differences in asthma burden persist across states even after accounting for air pollution. Some states, such as Wisconsin and Iowa, have significantly lower odds of elevated asthma burden, while others, such as New York, have significantly higher odds. For example, at the same AQI level, counties in Wisconsin have substantially lower odds of elevated asthma ED visits compared to the reference state. These findings indicate that geographic factors beyond air pollution play an important role in explaining asthma burden.

Predicted Risk Across AQI Levels. Predicted probabilities of elevated asthma ED status increase with AQI across all states (Figure 12), demonstrating a consistent positive relationship between air pollution and asthma risk. The curves are upward sloping and roughly parallel, indicating that the effect of AQI is similar across states. However, the curves are vertically separated, with states like New York consistently showing higher predicted risk and states like Wisconsin showing lower risk. This suggests that while air pollution increases asthma risk everywhere, state-level factors determine the overall level of risk.



- Curves are upward sloping → higher AQI → higher risk
- Curves are roughly parallel → AQI effect is similar across states
- Curves are vertically separated → states differ in baseline risk

4. Discussion

This study demonstrates that the apparent inverse association between air pollution exposure and asthma burden at the county level is not robust after accounting for geographic variation. While unadjusted analyses suggested a statistically significant inverse relationship, this association disappeared after adjusting for state, indicating strong confounding. In other words, the initial result was misleading and driven by differences across states rather than a true effect of air pollution.

From a methodological perspective, this highlights the limitation of relying on unadjusted contingency table analyses in ecological data. The change in results after adjustment reflects classical confounding, where overall (marginal) associations differ from associations observed after accounting for important factors.

A likely explanation is that counties with higher AQI are often more urban and have better access to healthcare, leading to improved asthma management and fewer emergency visits. In contrast, rural counties may have lower AQI but limited access to routine care, resulting in greater reliance on emergency services.

An important contribution of this study is showing that although including state as fixed effects helps control for confounding, it may also hide important geographic patterns. More appropriate approaches, such as multilevel (hierarchical) models or spatial models, could better capture these relationships by accounting for clustering and geographic dependence.

Several limitations should be noted. The ecological design means results cannot be applied to individuals. In addition, important factors such as socioeconomic status and healthcare access were not included, and spatial dependence between counties was not explicitly modeled, which may violate the assumption of independent observations.

5. Conclusion

In conclusion, the relationship between air pollution (AQI) and asthma burden at the county level is strongly influenced by geographic differences. While early analyses suggested that higher AQI was linked to lower asthma burden, this pattern did not hold after adjusting for state. Once these geographic differences were accounted for, the association weakened and was no longer statistically significant in most models, showing that the initial result was driven by confounding rather than a true effect. At the same time, the analysis showed that AQI still contributes some additional information beyond state and is positively related to asthma risk within states. However, state-level factors play a much larger role in determining overall asthma burden, as seen in the clear differences across states even after adjustment.

Overall, these findings highlight the importance of using appropriate statistical models when working with geographic data. Ignoring factors like state can lead to misleading conclusions. Future studies should use methods such as multilevel or spatial models to better capture geographic structure and provide more accurate insights into how environmental exposures affect health.

6. References

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7. Tables and Figures

Table 5. Pearson Chi Square Test (Association Between AQI Category and Asthma ED Status)

Table 5. Chi-Square Test of Association Between AQI Category and Asthma ED Status

Chi-Square Statistic	Degrees of Freedom	p-value
22.61	2	<0.001

Table 6. Likelihood Ratio (G^2) Test (Association Between AQI Category and Asthma ED Status)

Table 6. Likelihood Ratio Test (G-Test) of Association Between AQI Category and Asthma ED Status

G Statistic	Degrees of Freedom	p-value
22.74	2	<0.001

Table 7. Cochran-Armitage Trend Test (Association Between AQI Category and Asthma ED Status)

Table 7. Cochran–Armitage Trend Test Across AQI Categories and Asthma ED Status

Z Statistic	p-value
4.73	<0.001

Table 8. Contingency Table (Association Between Pollutant Category and Asthma ED Status)

Table 8. Association Between Pollutant Category and Asthma ED Status

	Asthma_ED_Status		
Pollutant Category	Not Elevated	Elevated	**Total**
Pollutant_Category			
Ozone	90 (43%)	118 (57%)	208 (100%)
PM2.5	125 (41%)	179 (59%)	304 (100%)
PM10	9 (56%)	7 (44%)	16 (100%)
Total	224 (42%)	304 (58%)	528 (100%)

Table 9. Pearson Chi Square Test (Association Between Pollutant Category and Asthma ED Status)

Table 9. Chi-Square Test of Association Between Pollutant Category and Asthma ED Status

Chi-Square Statistic	Degrees of Freedom	p-value
1.53	2	0.466

Table 10. AQI Category and Asthma ED Status, Stratified Analysis By Sex

Table 10. AQI Category and Asthma ED Status, Stratified by Sex		
AQI_Category	Not Elevated	Elevated
Female		
Low	30	66
Moderate	35	44
High	54	35
Male		
Low	30	66
Moderate	30	49
High	45	44

Table 11. Conchran-Mantel-Haenszel Test of Association Between AQI Category and Asthma ED Status, Stratified Analysis By Sex

Table 11. Cochran–Mantel–Haenszel Test of Association Between AQI Category and Asthma ED Status, Adjusted for Sex		
CMH Chi-Square	Degrees of Freedom	p-value
22.59	2	<0.001

Table 13. Odds Ratios for Asthma ED Status by AQI Category (Crude and Adjusted for State)

Table 13. Odds Ratios for Asthma ED Status by AQI Category (Crude and Adjusted for State)		
Model	OR (95% CI)	p-value
High vs Low		
Adjusted	1.41 (0.72, 2.78)	0.311
Crude	0.36 (0.24, 0.55)	<0.001
Moderate vs Low		
Adjusted	1.53 (0.83, 2.82)	0.173
Crude	0.65 (0.42, 1.01)	0.055

Table 14. Logistic Regression (Annual Weighted AQI)

Table 14. Logistic Regression Results: Continuous AQI Model							
Term	Coefficient	Std. Error	z value	p-value	OR	2.5% (OR)	97.5% (OR)
(Intercept)	0.5901	0.4104	1.4378	0.150	1.8042	0.8085	4.0520
Annual_Weighted_AQI	-0.0068	0.0096	-0.7110	0.477	0.9932	0.9746	1.0121

Table 17. Multiple Logistic Regression Models

Base Model: Annual Weighted AQI and State

Term	Base_Coefficient	Base_SE	Base_z	Base_p	Base_LCI_log	Base_UCI_log	Base_OR	Base_LCI_OR
(Intercept)	-1.2418	0.6848	-1.8135	0.07	-2.6065	0.0910	0.2889	0.0738
Annual_Weighted_AQI	0.0279	0.0126	2.2173	0.027	0.0035	0.0529	1.0283	1.0035
StateFlorida	4.5133	1.0873	4.1508	<0.001	2.7630	7.4642	91.2203	15.8473
StateIndiana	-0.3897	0.4745	-0.8213	0.412	-1.3271	0.5454	0.6773	0.2652
StateIowa	-0.6526	0.5484	-1.1900	0.234	-1.7463	0.4159	0.5207	0.1744
StateKansas	0.5977	0.6007	0.9950	0.32	-0.5686	1.8044	1.8179	0.5663
StateNebraska	-0.2795	0.6332	-0.4415	0.659	-1.5494	0.9554	0.7561	0.2124
StateNew Jersey	1.9085	0.6710	2.8444	0.004	0.6609	3.3385	6.7430	1.9365
StateNew York	1.7073	0.5431	3.1434	0.002	0.6611	2.8045	5.5140	1.9369
StateNorth Carolina	0.2796	0.4728	0.5914	0.554	-0.6488	1.2175	1.3226	0.5227
StateOregon	-1.6995	0.5974	-2.8449	0.004	-2.9315	-0.5638	0.1828	0.0533
StateRhode Island	1.7689	1.1697	1.5123	0.13	-0.2396	4.8005	5.8644	0.7869
StateSouth Carolina	2.6699	0.8389	3.1826	0.002	1.1961	4.6317	14.4383	3.3071
StateWisconsin	-1.2820	0.5106	-2.5109	0.012	-2.3030	-0.2887	0.2775	0.1000
Pollutant_CategoryPM2.5	NA	NA	NA	NA	NA	NA	NA	NA
Pollutant_CategoryPM10	NA	NA	NA	NA	NA	NA	NA	NA

Expanded Model: Annual Weighted AQI and State and Pollutant Category

Expanded_Coefficient	Expanded_SE	Expanded_z	Expanded_p	Expanded_LCI_log	Expanded_UCI_log	Expanded_OR	Expanded_LCI_OR	Expanded_UCI_OR
-0.6912	0.8228	-0.8400	0.401	-2.3153	0.9270	0.5010	0.0987	2.5269
0.0170	0.0154	1.1005	0.271	-0.0132	0.0476	1.0171	0.9869	1.0487
4.2395	1.1098	3.8200	<0.001	2.4250	7.2137	69.3733	11.3027	1357.9678
-0.6193	0.5125	-1.2084	0.227	-1.6354	0.3880	0.5383	0.1949	1.4741
-0.8897	0.5818	-1.5293	0.126	-2.0492	0.2438	0.4108	0.1288	1.2761
0.4407	0.6190	0.7121	0.476	-0.7630	1.6823	1.5539	0.4663	5.3779
-0.5420	0.6713	-0.8073	0.42	-1.8908	0.7636	0.5816	0.1510	2.1460
1.6427	0.7052	2.3293	0.02	0.3188	3.1289	5.1690	1.3754	22.8481
1.4458	0.5875	2.4610	0.014	0.3041	2.6233	4.2453	1.3553	13.7816
-0.0104	0.5304	-0.0196	0.984	-1.0584	1.0361	0.9897	0.3470	2.8183
-2.0949	0.6799	-3.0812	0.002	-3.4791	-0.7915	0.1231	0.0308	0.4532
1.3425	1.2197	1.1007	0.271	-0.7871	4.4301	3.8285	0.4552	83.9375
2.3693	0.8748	2.7084	0.007	0.8076	4.3798	10.6902	2.2425	79.8214
-1.4939	0.5416	-2.7584	0.006	-2.5775	-0.4409	0.2245	0.0760	0.6435
0.2981	0.2876	1.0368	0.3	-0.2653	0.8647	1.3473	0.7670	2.3743
-0.3990	0.6483	-0.6153	0.538	-1.7234	0.8523	0.6710	0.1785	2.3451

Alternative Model: AQI Category and State

Table 17. Logistic Regression Results: AQI Category + State (Alternative Model)									
Term	Coefficient	Std. Error	z value	p-value	2.5% (log-odds)	97.5% (log-odds)	OR	2.5% (OR)	97.5% (OR)
(Intercept)	-0.2370	0.4345	-0.5455	0.585	-1.1015	0.6186	0.7890	0.3324	1.8563
AQI_CategoryModerate	0.4227	0.3100	1.3633	0.173	-0.1813	1.0370	1.5260	0.8342	2.8208
AQI_CategoryHigh	0.3469	0.3425	1.0129	0.311	-0.3219	1.0241	1.4147	0.7248	2.7846
StateFlorida	4.5174	1.0926	4.1345	<0.001	2.7518	7.4736	91.5960	15.6706	1760.9715
StateIndiana	-0.5607	0.4807	-1.1663	0.244	-1.5132	0.3833	0.5708	0.2202	1.4672
StateIowa	-0.5985	0.5469	-1.0944	0.274	-1.6879	0.4685	0.5496	0.1849	1.5976
StateKansas	0.5206	0.5949	0.8752	0.382	-0.6356	1.7146	1.6831	0.5296	5.5545
StateNebraska	-0.5305	0.6284	-0.8441	0.399	-1.7948	0.6910	0.5883	0.1662	1.9956
StateNew Jersey	1.7220	0.6757	2.5487	0.011	0.4623	3.1584	5.5959	1.5877	23.5339
StateNew York	1.4456	0.5285	2.7354	0.006	0.4237	2.5101	4.2442	1.5276	12.3061
StateNorth Carolina	0.2102	0.4717	0.4457	0.656	-0.7178	1.1435	1.2340	0.4878	3.1377
StateOregon	-1.7878	0.5921	-3.0195	0.002	-3.0116	-0.6650	0.1673	0.0492	0.5143
StateRhode Island	1.4495	1.1700	1.2389	0.215	-0.5616	4.4811	4.2609	0.5703	88.3291
StateSouth Carolina	2.6961	0.8430	3.1981	0.001	1.2127	4.6636	14.8222	3.3625	106.0194
StateWisconsin	-1.3649	0.5268	-2.5907	0.01	-2.4181	-0.3406	0.2554	0.0891	0.7113

Table 26. State-Level Effects on Asthma ED Status (Adjusted for Annual Weighted AQI)

Table 26. State-Level Effects on Asthma ED Status (Adjusted for Annual Weighted AQI)

States ranked by log-odds (lowest to highest relative to reference state)

State	Log-Odds	Std. Error	p-value
Oregon	-19.760	1769.105	0.991
Wisconsin	-1.637	0.539	0.002
Iowa	-1.212	0.592	0.041
Indiana	-0.679	0.501	0.176
Nebraska	-0.637	0.668	0.340
North Carolina	0.100	0.493	0.840
Kansas	0.688	0.658	0.296
New York	1.595	0.561	0.005
New Jersey	19.335	2087.261	0.993
South Carolina	19.435	2074.495	0.993
Rhode Island	19.515	4780.413	0.997
Florida	19.603	1218.231	0.987

Figure 9. Receiver Operating Characteristic (ROC) Curve with Influential Observations, included

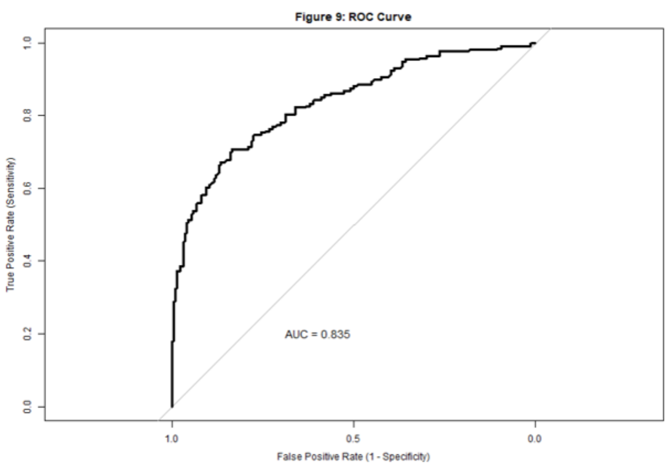
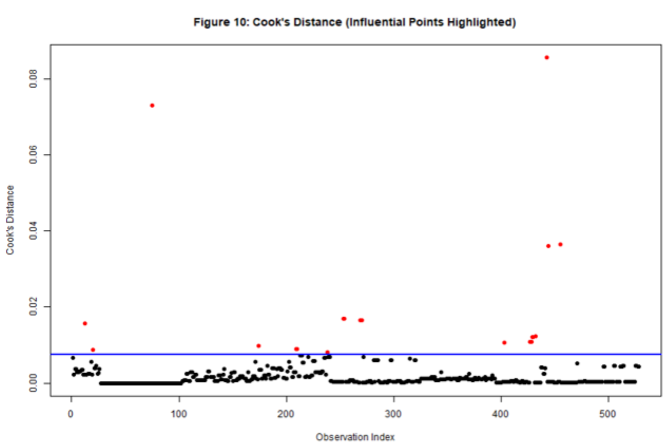


Figure 10. Cook's Distance (Influential Observations Highlighted_



Note: Figure numbering begins at Figure 2. There is no Figure 1.

R Packages Used:

readxl, dplyr, ggplot2, gtsummary, gt, vcd, tidyr, scales, DescTools, broom, MASS, ResourceSelection, pROC